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Guo et al.

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(54) **FIELD EFFECT TRANSISTOR WITH
SELECTIVE CHANNEL LAYER DOPING**

(71) Applicant: **Wolfspeed, Inc.**, Durham, NC (US)

(72) Inventors: **Jia Guo**, Apex, NC (US); **Saptharishi
Sriram**, Cary, NC (US); **Scott
Sheppard**, Chapel Hill, NC (US)

(73) Assignee: **Wolfspeed, Inc.**, Durham, NC (US)

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See application file for complete search history.

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Primary Examiner — Eva Y Montalvo

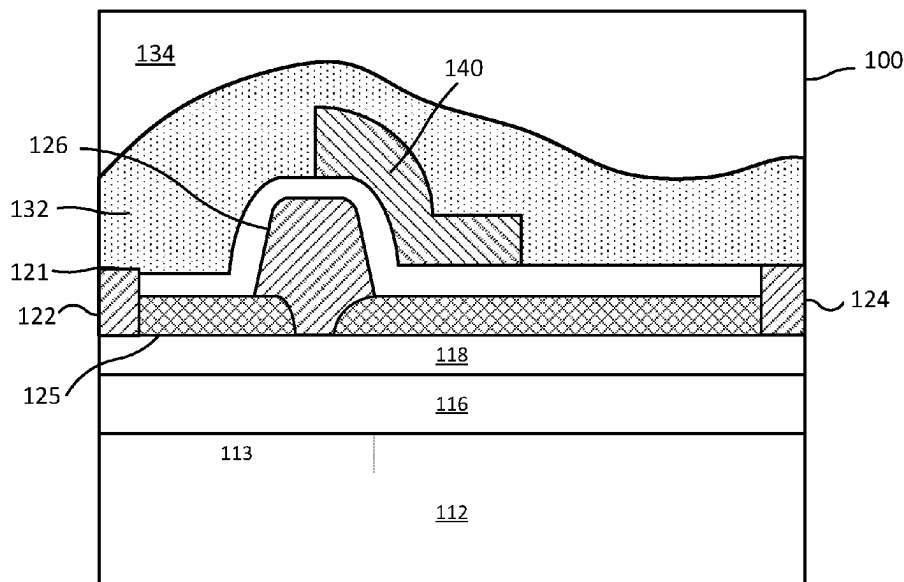
Assistant Examiner — Pratiksha Jayant Lohakare

(74) *Attorney, Agent, or Firm* — Sage Patent Group

(57) **ABSTRACT**

A transistor device according to some embodiments includes
a channel layer, a barrier layer on the channel layer, and
source and drain contacts on the barrier layer, and a gate
contact on the barrier layer between the source and drain
contacts. The channel layer includes a sub-layer having an
increased doping concentration level relative to a remaining
portion of the channel layer. The presence of the sub-layer
may reduce drain lag without substantially increasing gate
lag.

21 Claims, 14 Drawing Sheets



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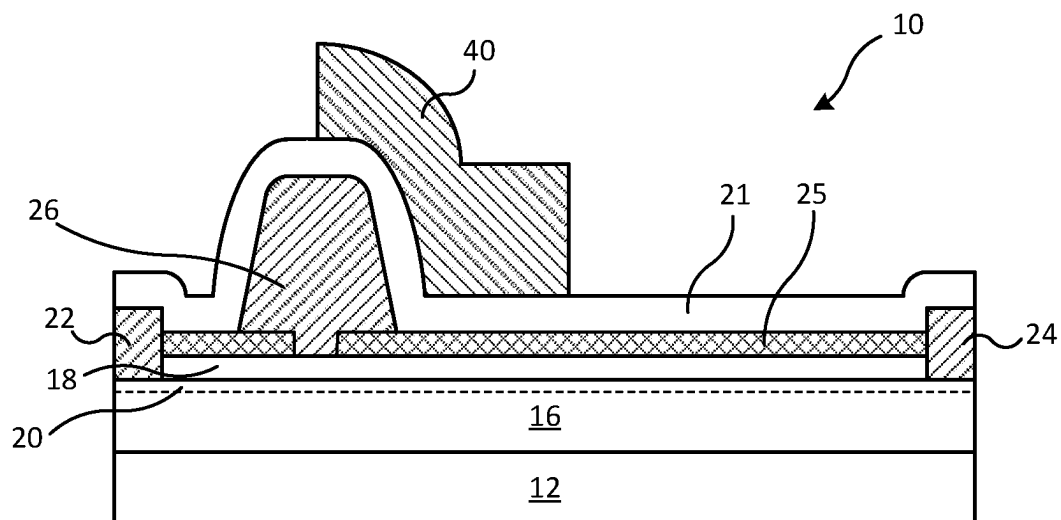


FIGURE 1
PRIOR ART

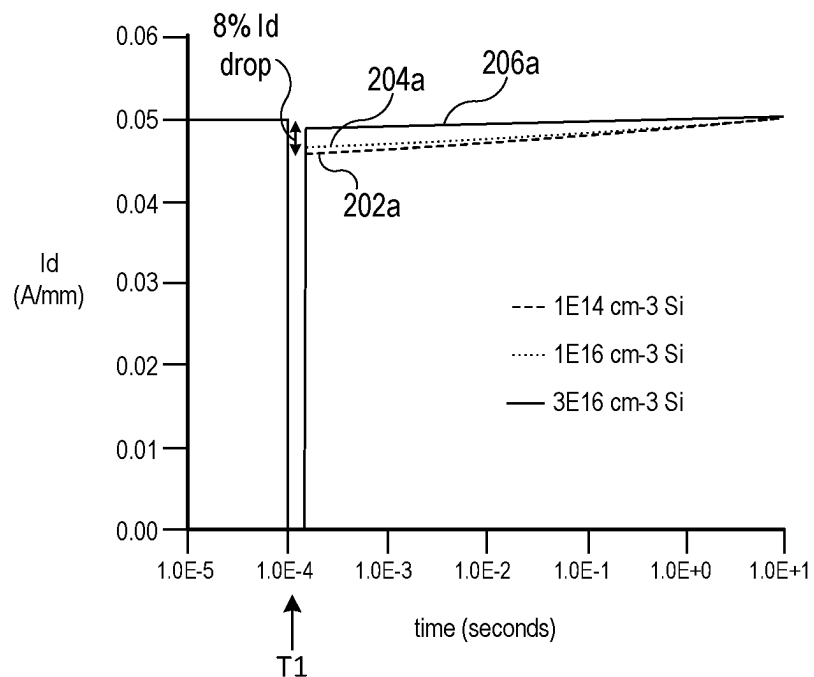


FIGURE 2A

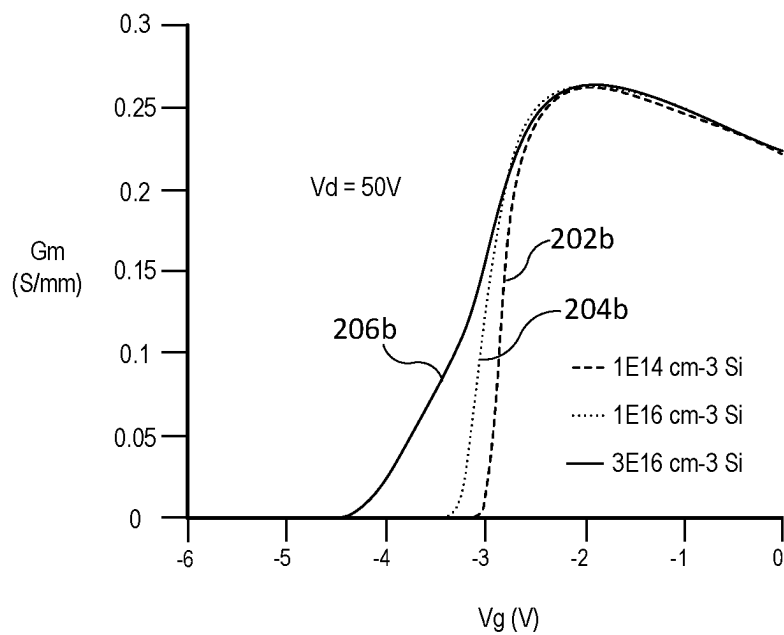


FIGURE 2B

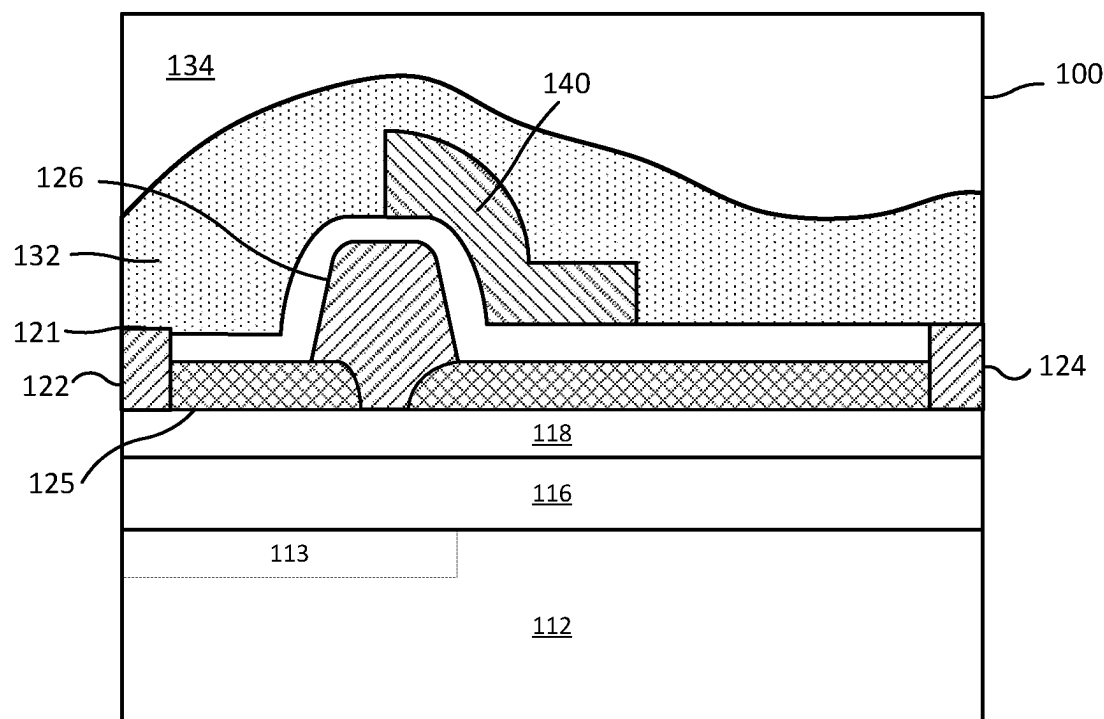


FIGURE 3

Type A 116A	Type B 116B	Type C 116C
Increased doping <u>116-3</u>	Increased doping <u>116-3</u>	Lightly doped <u>116-3</u>
Increased doping <u>116-2</u>	Lightly doped <u>116-2</u>	Increased doping <u>116-2</u>
Lightly doped <u>116-1</u>	Lightly doped <u>116-1</u>	Lightly doped <u>116-1</u>

FIGURE 4

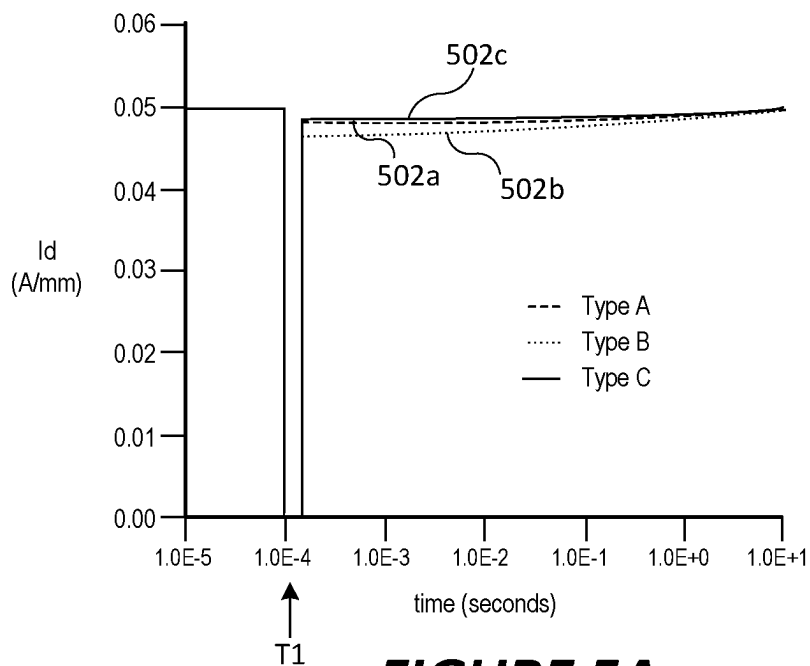


FIGURE 5A

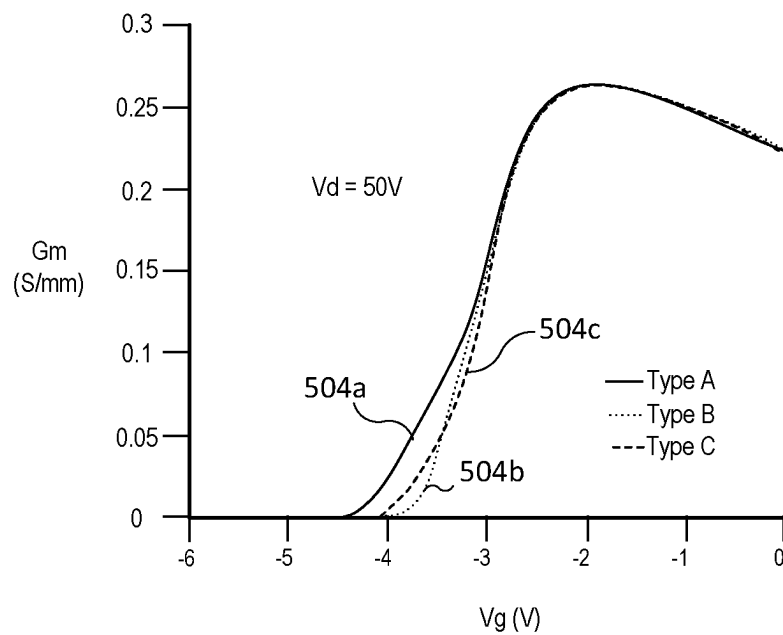


FIGURE 5B

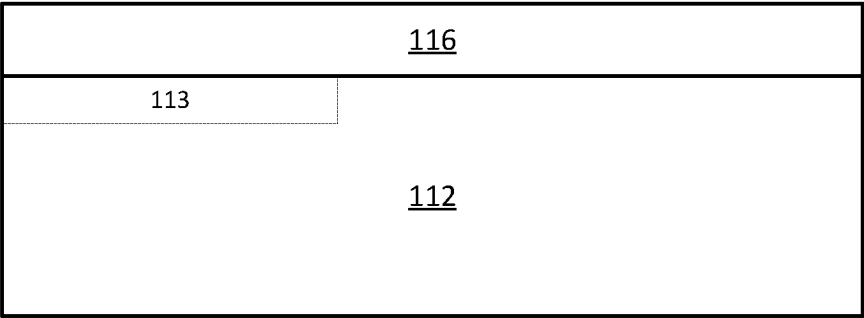


FIGURE 6A

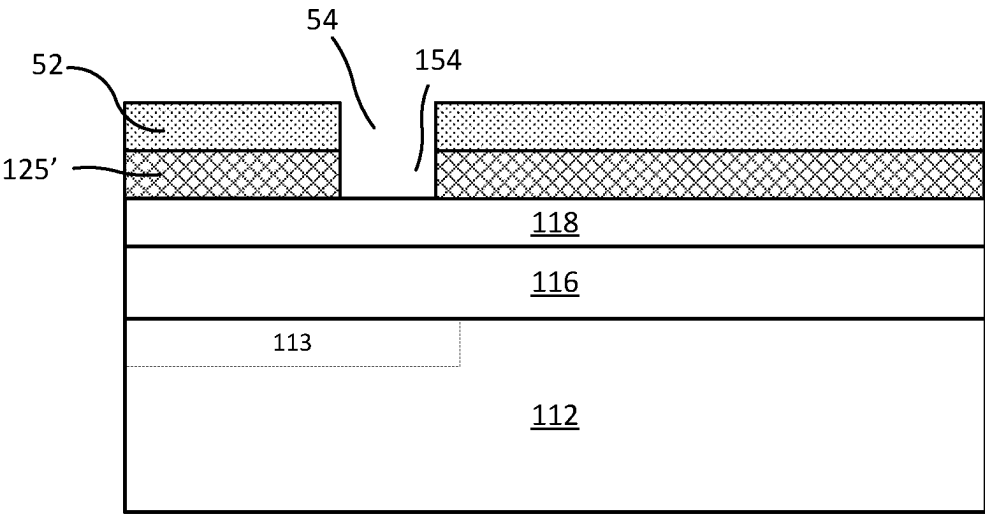


FIGURE 6B

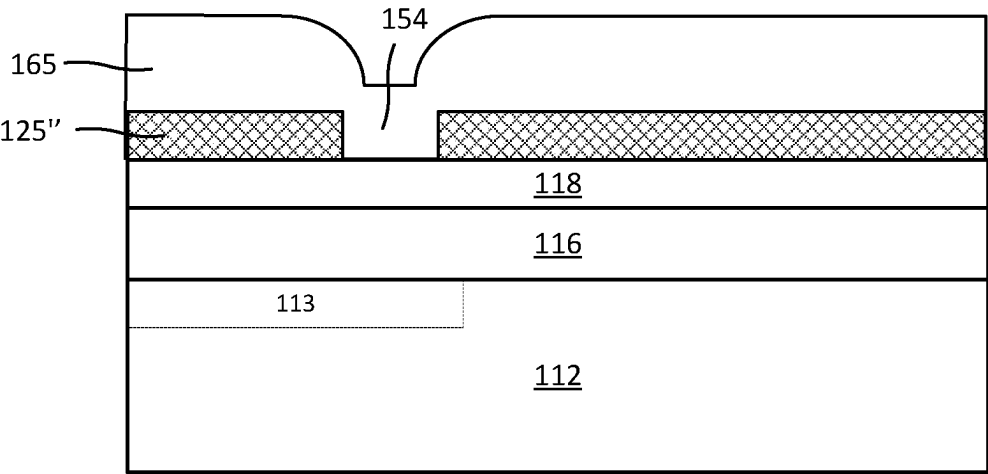


FIGURE 6C

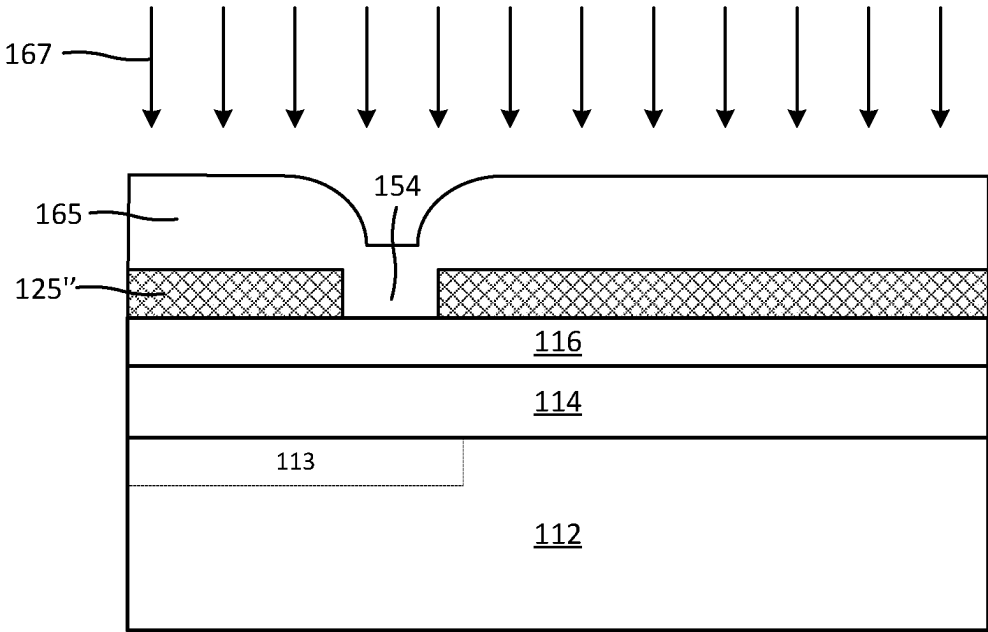


FIGURE 6D

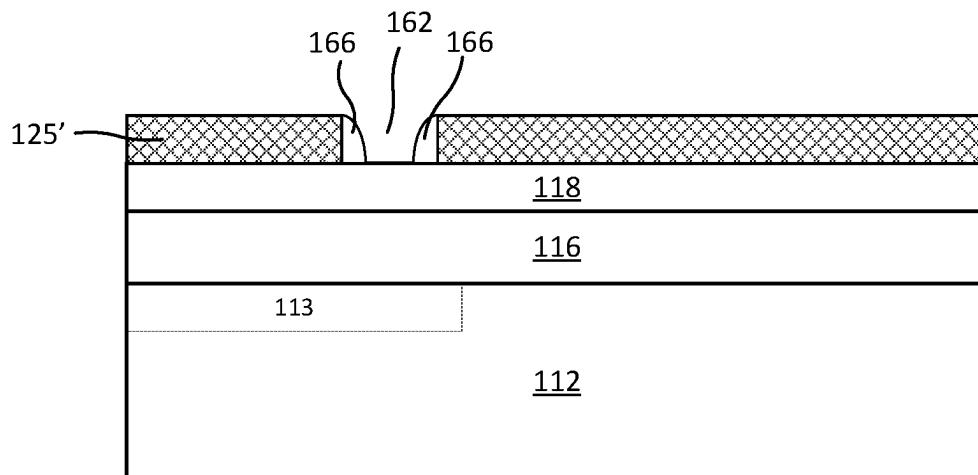


FIGURE 6E

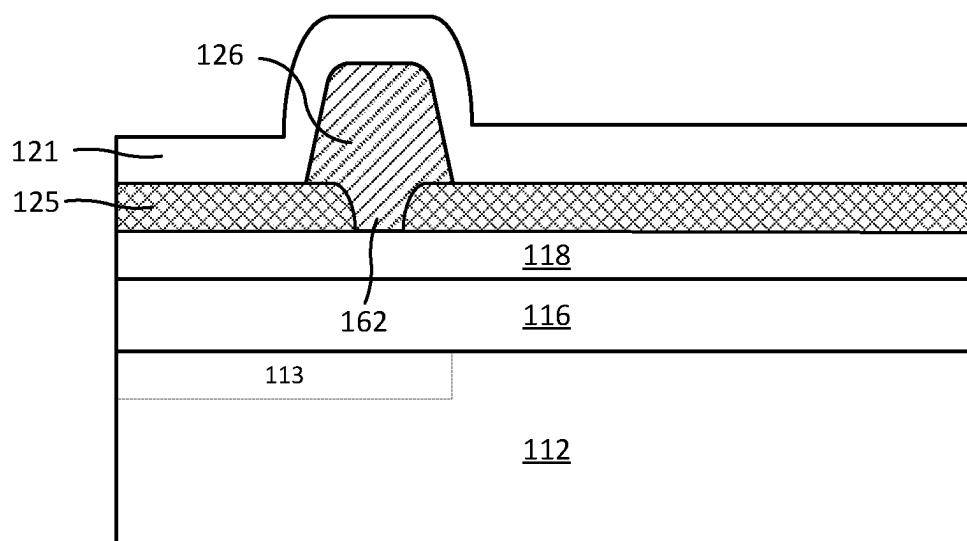


FIGURE 6F

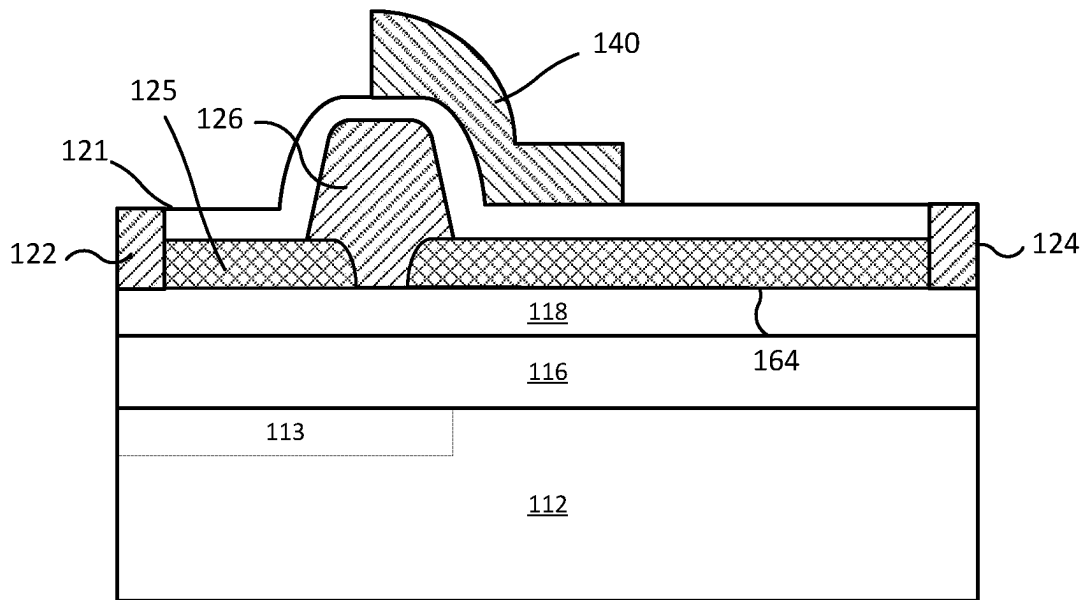


FIGURE 6G

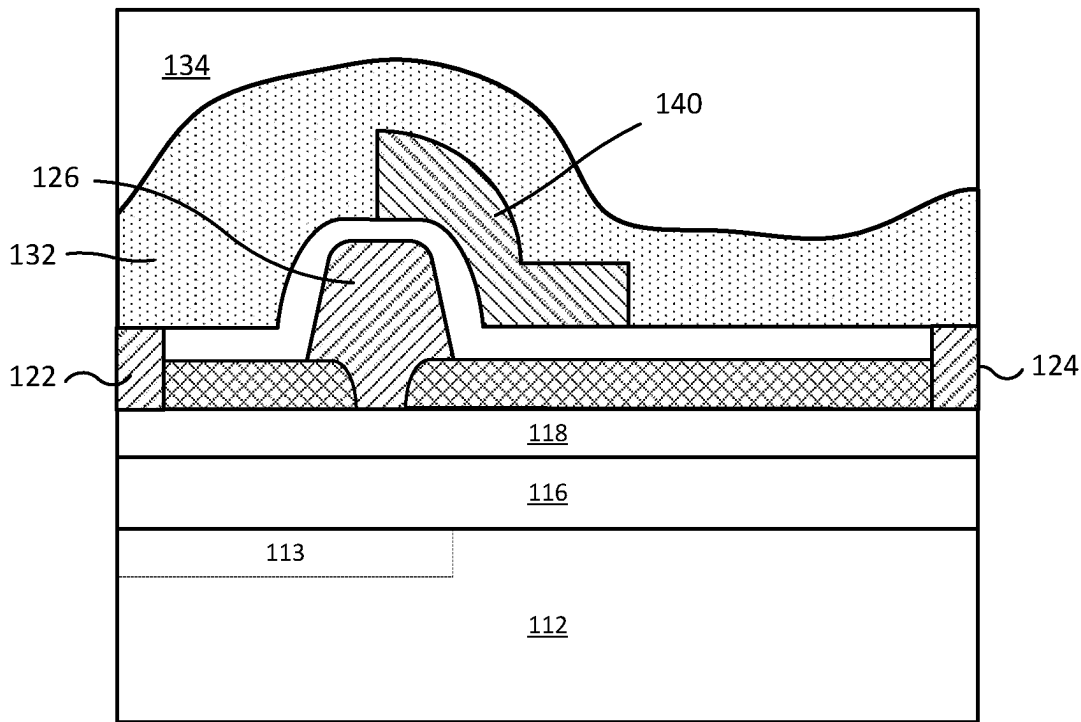
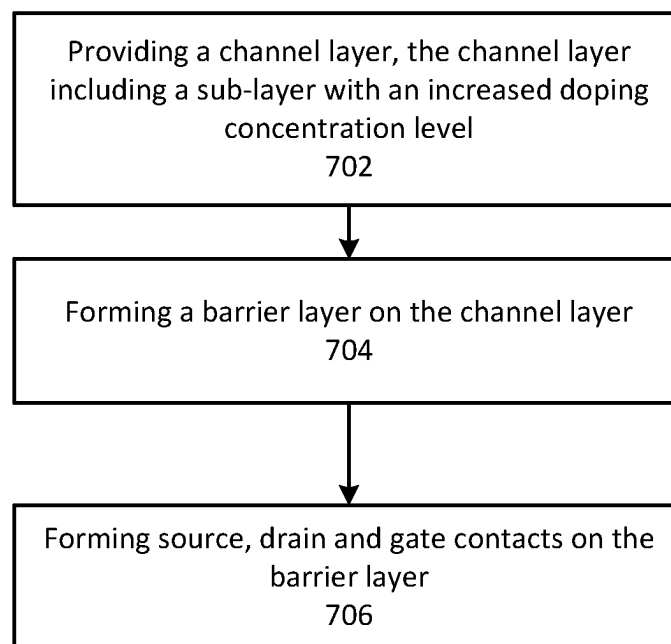


FIGURE 6H

**FIGURE 7**

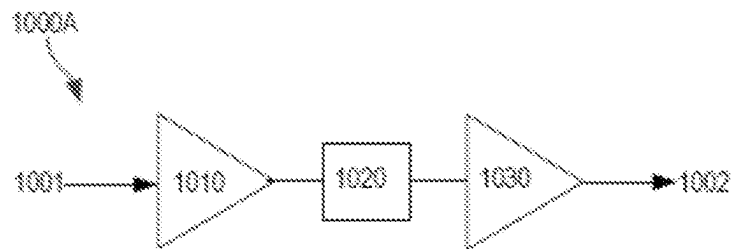


FIG. 8A

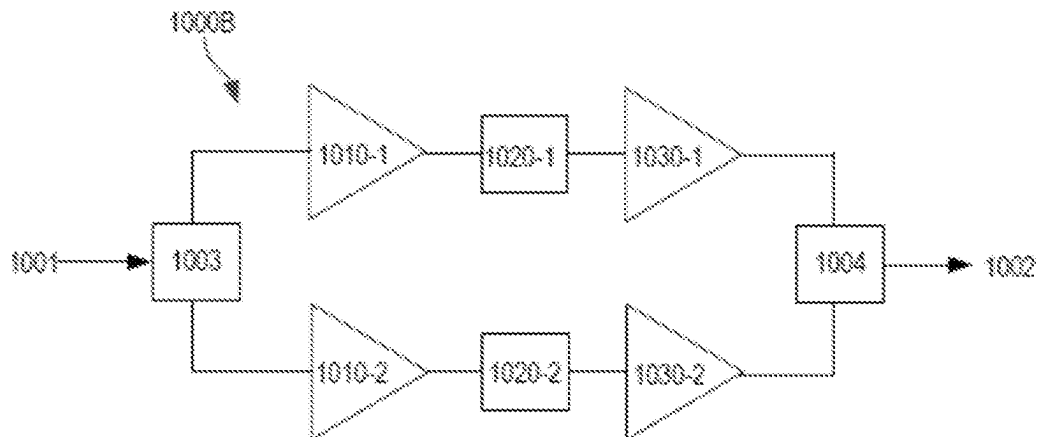


FIG. 8B

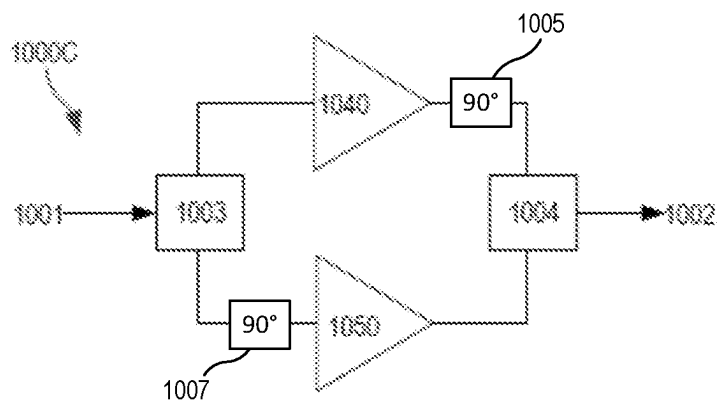


FIG. 8C

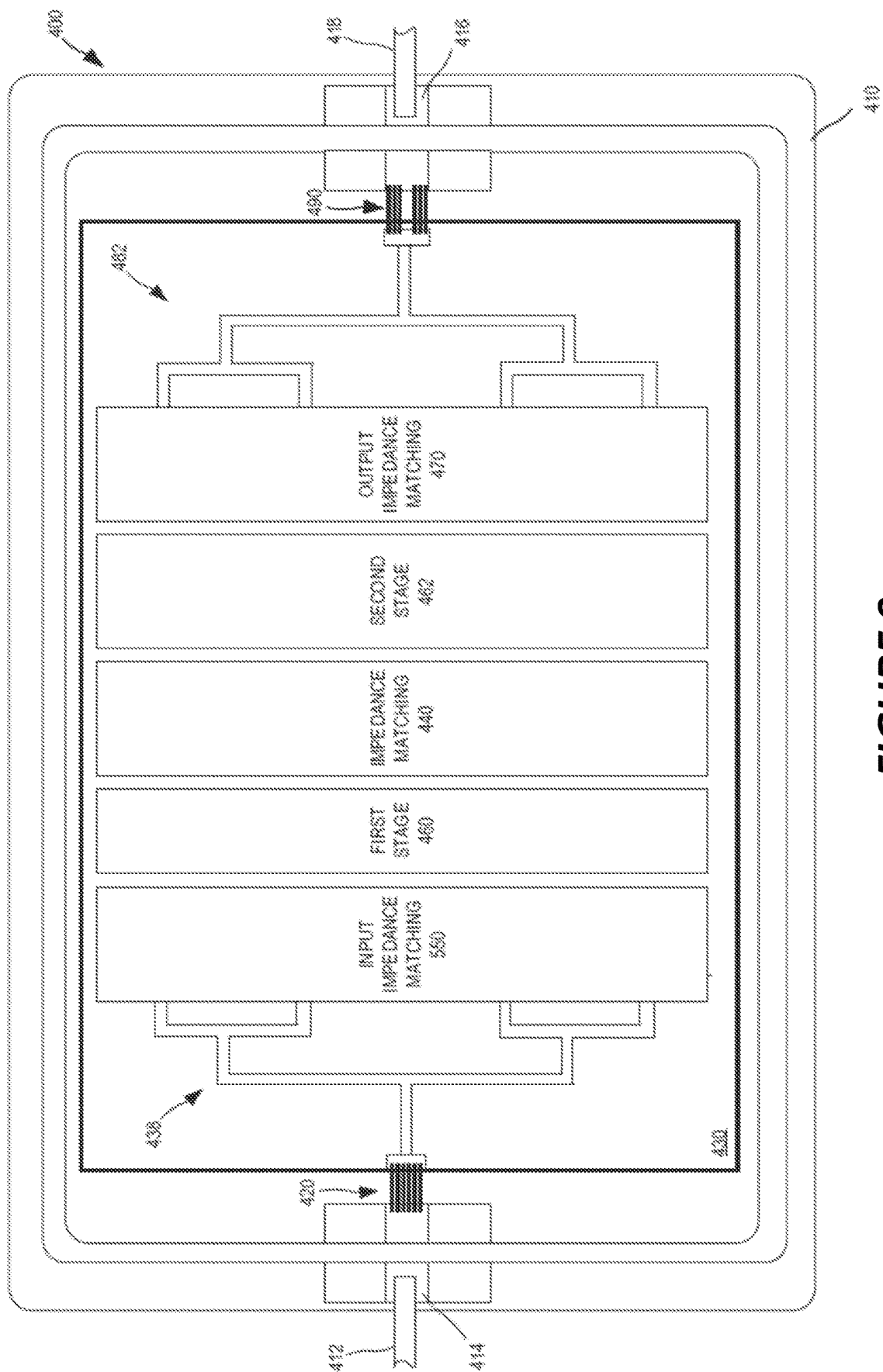


FIGURE 9

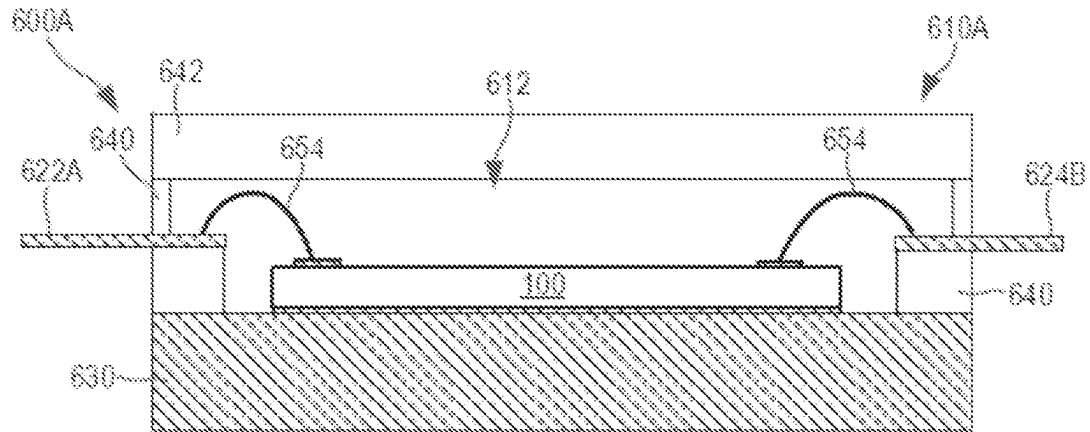


FIGURE 10A

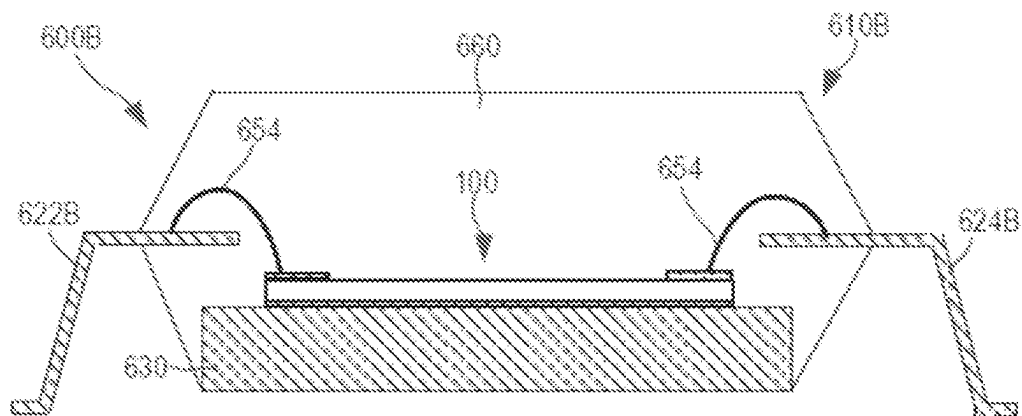


FIGURE 10B

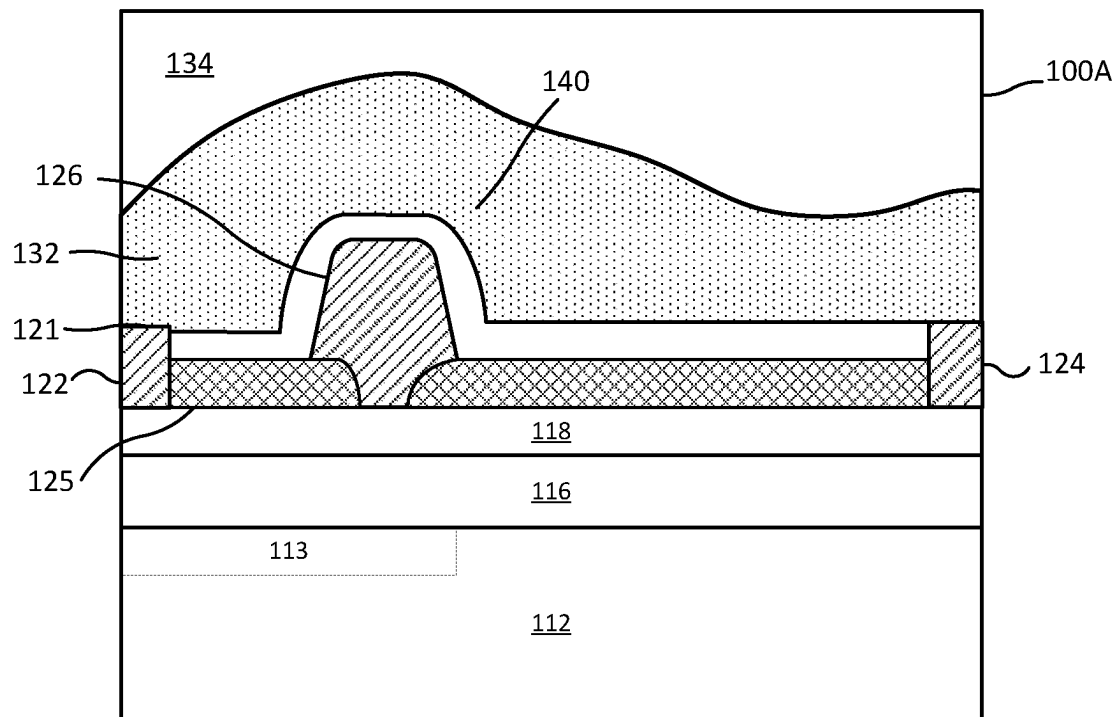


FIGURE 11A

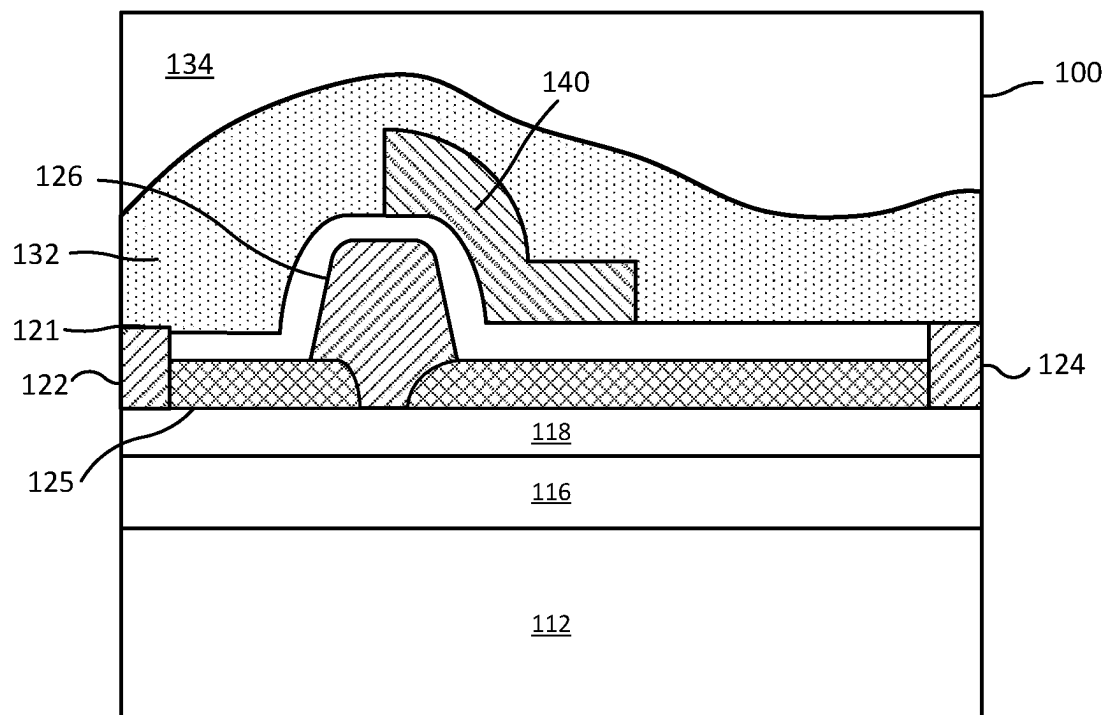


FIGURE 11B

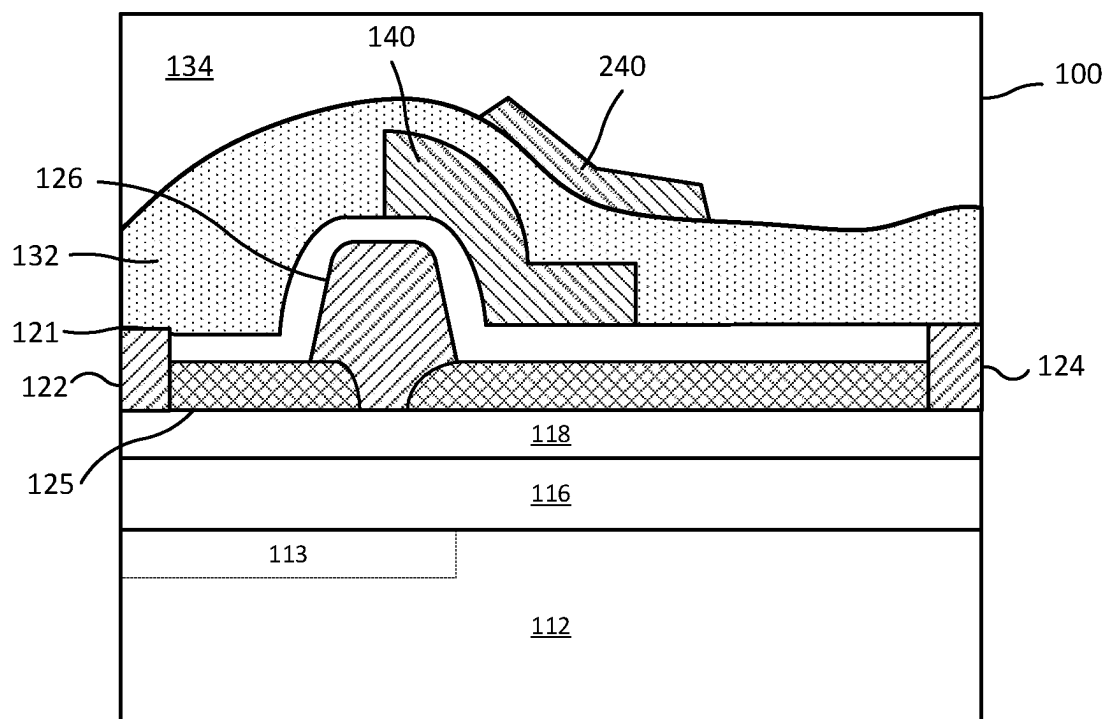


FIGURE 11C

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FIELD EFFECT TRANSISTOR WITH SELECTIVE CHANNEL LAYER DOPING

STATEMENT OF GOVERNMENT RIGHTS

This invention was made with Government support under Department of Defense Contract No. N000164-19-C-WP50 awarded by the Department of the Navy. The Government has certain rights in the invention.

BACKGROUND

The present disclosure relates to transistor structures and in particular to GaN-based high electron mobility transistors.

Narrow bandgap semiconductor materials, such as silicon (Si) and gallium arsenide (GaAs), are widely used in semiconductor devices for low power and, in the case of Si, low frequency applications. However, these semiconductor materials may not be well-suited for high power and/or high frequency applications, for example, due to their relatively small bandgaps (1.12 eV for Si and 1.42 for GaAs at room temperature) and relatively small breakdown voltages.

Interest in high power, high temperature and/or high frequency applications and devices has focused on wide bandgap semiconductor materials such as silicon carbide (3.2 eV for 4H—SiC at room temperature) and the Group III nitrides (e.g., 3.36 eV for GaN at room temperature). These materials may have higher electric field breakdown strengths and higher electron saturation velocities than GaAs and Si.

A device of particular interest for high power and/or high frequency applications is the High Electron Mobility Transistor (HEMT), which is also known as a modulation doped field effect transistor (MODFET). In a HEMT device, a two-dimensional electron gas (2DEG) may be formed at the heterojunction of two semiconductor materials with different bandgap energies, where the smaller bandgap material has a higher electron affinity than the wider bandgap material. The 2DEG is an accumulation layer in the undoped smaller bandgap material and can contain a relatively high sheet electron concentration, for example, in excess of 10^{13} carriers/cm². Additionally, electrons that originate in the wider bandgap semiconductor may transfer to the 2DEG, allowing a relatively high electron mobility due to reduced ionized impurity scattering. This combination of relatively high carrier concentration and carrier mobility can give the HEMT a relatively large transconductance and may provide a performance advantage over metal-semiconductor field effect transistors (MESFETS) for high-frequency applications.

HEMTs fabricated in the gallium nitride/aluminum gallium nitride (GaN/AlGaN) material system can generate large amounts of RF power due to a combination of material characteristics, such as relatively high breakdown fields, relatively wide bandgaps, relatively large conduction band offset, and/or relatively high saturated electron drift velocity. A major portion of the electrons in the 2DEG may be attributed to polarization in the AlGaN.

Field plates have been used to enhance the performance of GaN-based HEMTs at microwave frequencies and have exhibited performance improvement over devices without field plates. Many field plate approaches have involved a field plate connected to the source of the transistor with the field plate on top of the drain side of a channel. This can result in a reduction of the electric field on the gate-to-drain side of the transistor, thereby increasing breakdown voltage and reducing the high-field trapping effect. However, some

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transistors with gate-to-drain field plates can exhibit relatively poor reliability performance, particularly at class C (or higher class) operation where the electric field on the source side of the gate becomes significant.

FIG. 1 shows a GaN-based HEMT transistor device 10 formed on a silicon carbide substrate 12. A GaN channel layer 16 (also referred to as a GaN buffer layer 16) is on the substrate 12, and an AlGaN barrier layer 18 is on the channel layer 16. A two-dimensional electron gas (2DEG) 20 arises in the channel layer 16 adjacent the barrier layer 18. A source contact 22 and a drain contact 24 are formed on the channel layer 16. The conductivity of the 2DEG 20 is modulated by applying a voltage to a gate 26 that is formed on the barrier layer 18 between the source contact 22 and the drain contact 24. As shown in FIG. 1, the gate 26 may have a mushroom or T-top configuration in which the gate 26 contacts the barrier layer 18 in a relative narrow contact region that extends through a surface dielectric layer 25.

The transistor device 10 includes a field plate 40 that is connected to the source contact 22. The field plate 40 is spaced apart from the gate 26 by an interlayer dielectric layer 21, and is spaced apart from the barrier layer 18 by the interlayer dielectric layer 21 and the surface dielectric layer 25. The field plate 40 extends above the gate 26 and laterally toward the drain contact 24.

The field plate 40 is connected to the source contact 22. Connecting the field plate 40 to the source contact 22 provides a reduction in gate-to-drain capacitance (Cgd), which consequently can enhance the gain of the device. In addition to reducing gate-to-drain capacitance Cgd, the presence of the field plate 40 may improve linearity of the device and/or reduce the drain bias dependence of the capacitance. While GaN-based HEMTs generally display good linearity, further improvement may be desired for in high power RF applications. Moreover, while the structure shown in FIG. 1 can have a reduced gate-to-drain capacitance Cgd compared to structures without a field plate, the gate-to-drain capacitance Cgd can still show a large dependence on the bias of the drain contact 24.

SUMMARY

A transistor device according to some embodiments includes a channel layer, a barrier layer on the channel layer, and source and drain contacts on the barrier layer, and a gate contact on the barrier layer between the source and drain contacts. The channel layer includes a sub-layer having an increased doping concentration level relative to a remaining portion of the channel layer.

The sub-layer may be a buried sub-layer, and the device may further include a second sub-layer between the buried sub-layer and the barrier layer. The buried sub-layer has a thickness between 50 and 150 nm, and in some embodiments a thickness of about 100 nm. The second sub-layer may have a thickness between 50 and 150 nm, and in some embodiments a thickness of about 100 nm.

The buried sub-layer may be doped with an n-type dopant, such as silicon, having a doping concentration level greater than 2×10^{16} cm⁻³.

The doping concentration level of the n-type dopant may be between about 2×10^{16} cm⁻³ and 4×10^{16} cm⁻³, and in some cases about 3×10^{16} cm⁻³. The remaining portions of the channel layer may be doped with an n-type dopant having a doping concentration level less than 2×10^{16} cm⁻³, such as about 1×10^{16} cm⁻³.

The sub-layer may be a second sub-layer having a second doping concentration level, and the channel layer may

include a first sub-layer having a first doping concentration level and a third sub-layer having a third doping concentration level. The second sub-layer is between the first sub-layer and the third sub-layer, and the second doping concentration level is higher than the first doping concentration level and the third doping concentration level. The second doping concentration level may be between about $2 \times 10^{16} \text{ cm}^{-3}$ and $4 \times 10^{16} \text{ cm}^{-3}$, and the first and third doping concentration levels may each be below about $2 \times 10^{16} \text{ cm}^{-3}$.

The channel layer may have a total thickness of about 500 nm or less.

The barrier layer and the channel layer may cooperate to induce a two-dimensional electron gas (2DEG) in the channel layer, and the transistor device may be a high electron mobility transistor. The channel layer may include GaN.

A method of forming a transistor device includes providing a channel layer, forming a barrier layer on the channel layer, and forming source and drain contacts on the barrier layer, and a gate contact on the barrier layer between the source and drain contacts. The channel layer may include a sub-layer having an increased doping concentration level relative to a remaining portion of the channel layer.

A transistor device according to further embodiments includes a channel layer, a barrier layer on the channel layer, and source and drain contacts on the barrier layer, and a gate contact on the barrier layer between the source and drain contacts. The channel layer includes a first sub-layer having a first doping concentration level, a second sub-layer having a second doping concentration level on the first sub-layer, and a third sub-layer having a third doping concentration level on the second sub-layer opposite the first sub-layer, wherein the third sub-layer is adjacent the barrier layer, and the second doping concentration level is higher than the first doping concentration level and the third doping concentration level.

The second doping concentration level may be between about $2 \times 10^{16} \text{ cm}^{-3}$ and $4 \times 10^{16} \text{ cm}^{-3}$, and the first and third doping concentration levels may each be less than about $2 \times 10^{16} \text{ cm}^{-3}$.

The first and third doping concentration levels may each be about $1 \times 10^{16} \text{ cm}^{-3}$, and the second doping concentration level may be about $3 \times 10^{16} \text{ cm}^{-3}$. The first and second sub-layers may each have a thickness between about 50 nm and 150 nm.

The transistor may further include a buried p-type region in the substrate beneath the source contact. In some embodiments, the transistor device may further include a field plate over the barrier layer between the gate contact and the drain contact, wherein the field plate is separated from the gate by an insulating layer. In some embodiments, the transistor device may further include a second field plate over the barrier layer between the gate contact and the drain contact.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a cross-sectional view of a conventional transistor device including a field plate.

FIG. 2A is a graph of drain current versus time for conventional device structures.

FIG. 2B is a graph of transconductance as a function of gate voltage for conventional device structures.

FIG. 3 is a cross-sectional view of a transistor device including a field plate in accordance with some embodiments.

FIG. 4 illustrates channel layer doping arrangements of a transistor device structure according to various embodiments.

FIG. 5A is a graph of drain current versus time for a device structure according to some embodiments.

FIG. 5B is a graph of transconductance as a function of gate voltage for a device structure according to some embodiments.

FIGS. 6A to 6H are cross-sectional views illustrating operations for manufacturing a transistor device including a field plate in accordance with some embodiments.

FIG. 7 is a block diagram illustrating operations of forming a transistor device according to some embodiments.

FIGS. 8A-8C are schematic block diagrams of multi-amplifier circuits in which RF transistor amplifiers incorporating transistor devices according to embodiments may be used.

FIG. 9 is a schematic plan view of a monolithic microwave integrated circuit RF transistor amplifier according to some embodiments.

FIGS. 10A and 10B are schematic cross-sectional views illustrating two example ways that an RF transistor device according to some embodiments may be packaged to provide RF transistor amplifiers.

FIGS. 11A to 11C are cross-sectional views of transistor devices in accordance with various further embodiments.

DETAILED DESCRIPTION OF EMBODIMENTS

Embodiments of the inventive concepts will now be described in connection with the accompanying drawings.

In conventional GaN HEMT devices, such as the transistor device 10 shown in FIG. 1, the channel layer 16 may be doped with a dopant, such as carbon, to provide electron traps that help to achieve a desired breakdown voltage. The electron traps may cause memory effects that adversely affect device performance, such as, for example, causing gate lag. Reduction of gate lag as a result of doping the channel layer with an n-type dopant, such as silicon, is illustrated, for example, in FIG. 2A, which is a graph of simulated values of drain current (I_d) as a function of time following a gate stress event for devices having uniformly doped channel layers with three different channel layer doping concentration levels, namely, $1 \times 10^{14} \text{ cm}^{-3}$ (curve 202a), $1 \times 10^{16} \text{ cm}^{-3}$ (curve 204a) and $3 \times 10^{16} \text{ cm}^{-3}$ (curve 206a). As shown in FIG. 2A, after a -12V gate stress event at time T1, the drain current in the device with a more heavily doped channel layer (206) suffers a lower I_d drop and recovers more quickly than in the devices with more lightly doped channel layers (202a, 204a).

However, devices with more heavily doped channel layers may suffer from soft pinch-off or turn-on, which is a phenomenon in which the drain current of the device responds more slowly to changes in gate voltage, which is undesirable. This effect is illustrated in FIG. 2B, which is a graph of simulated values of transconductance (G_m) as a function of gate voltage (V_g) for devices having three different channel layer doping concentration levels, namely, $1 \times 10^{14} \text{ cm}^{-3}$ (curve 202b), $1 \times 10^{16} \text{ cm}^{-3}$ (curve 204b) and $3 \times 10^{16} \text{ cm}^{-3}$ (curve 206b). As shown in FIG. 2B, the transconductance curve of the device having a more heavily doped channel layer (curve 206b) has a lower slope relative to the devices having more lightly doped channel layers. Thus, when doping the channel layer with uniform doping, there is an undesirable tradeoff between gate lag and pinch-off/turn-on, in that doping the channel layer to improve gate lag may result in slower response of the drain current to gate voltage.

Some embodiments described herein avoid this and other problems by providing non-uniform doping of a channel layer. In particular, some embodiments described herein

provide selectively doped buried layers in the channel layer with increased doping concentration levels. These layers may compensate the electron trapping effect so that gate lag may be reduced under high gate bias. With the proper doping profile, it has been found that the gate lag problem can be overcome without introducing other negative effects, such as soft pinch-off.

Referring to FIG. 3, a transistor device according to some embodiments is illustrated. In particular, FIG. 3 illustrates a high electron mobility transistor (HEMT) device **100** formed on a substrate **112**. A channel layer **116** is formed on the substrate **112**, and a barrier layer **118** is on the channel layer **116**.

A source contact **122** and a drain contact **124** are formed on the barrier layer **118**. A gate **126** is formed on the barrier layer **118** between the source contact **122** and drain contact **124**. As shown in FIG. 3, the gate **126** may have a mushroom or T-top configuration in which the gate **126** contacts the barrier layer **118** in a relatively narrow contact region of a recessed contact portion **127** of the gate **126** that extends through a surface dielectric layer **125** within a gate aperture **162** that extends through the surface dielectric layer **125**.

The source and drain contacts **122**, **124**, may include a metal, such as TiAlN, that can form an ohmic contact to Group III nitride-based materials. The gate **126** may include a metal capable of making a Schottky contact to a nitride based semiconductor material, such as Ni, Pt, NiSix, Cu, Pd, Cr, TaN, W, and/or WSiN.

A buried p-type region **113** is formed in the substrate **112** beneath the channel layer and may be connected to the source contact **122** of the device. The buried p-type region **113** may extend vertically to a depth of about 0.75 microns into the substrate, and may extend laterally past the gate contact **126** and toward the drain contact **124** by a distance of about 0.4 microns.

The buried p-type region **113** may be floating or may be connected to the source contact **122** or to the gate **126**, or may be independently biased. The buried p-type region **113** may be omitted in some embodiments. The buried p-type region **113** may be formed as described in U.S. Pat. No. 10,892,356, entitled "Group III-nitride high-electron mobility transistors with buried p-type layers and process for making the same," the disclosure of which is incorporated herein by reference in its entirety.

In some embodiments, the buried p-type region **113** may be formed via epitaxial growth, selective etching and epitaxial regrowth on the substrate and/or via ion implantation. The buried p-type region **113** may extend laterally beneath the source contact **122** and the gate contact **126**. In some embodiments, the buried p-type region **113** may be spaced laterally from the source contact **122** and/or the gate contact **126** so that it does not extend laterally beneath the source contact **122** and/or the gate contact **126**.

In some embodiments, the substrate **112** includes silicon carbide, the channel layer **116** includes GaN, and the barrier layer includes AlGaIn. However, it will be appreciated that other materials or combinations of materials can be used. Moreover, the channel layer **116** and/or the barrier layer **118** may include an alloy such as $\text{Al}_x\text{Ga}_{1-x}\text{N}$, where $0 \leq x \leq 1$. It will be further appreciated that although a HEMT device is illustrated, the HEMT device **100** may be another type of transistor device, such as a metal-semiconductor field effect transistor (MESFET), a junction field effect transistor (JFET), a metal oxide semiconductor field effect transistor (MOSFET), etc.

The HEMT device **100** may include a field plate **140** that is connected to the source contact **122** via a connection

outside the active region of the device. The field plate **140** is spaced apart laterally from the gate **126** by an interlayer dielectric layer **121**. The field plate **140** may be electrically connected to the source contact **122** outside the active region of the device (and outside the plane illustrated in FIG. 3). In some embodiments, the field plate **140** may be connected to the gate **126** instead of the source contact **122**, or may be independently biased.

The field plate **140** is generally vertically spaced apart from the barrier layer **118** by the interlayer dielectric layer **121** and the surface dielectric layer **125**.

It will be appreciated that in some embodiments, the field plate **140** may be omitted.

A passivation layer **132** is formed over the interlayer dielectric layer **121** and the field plate **140**, and a field dielectric layer **134** is formed over the passivation layer **132**. The passivation layer **132** may fill a gap between the field plate **140** and the gate **126** that is not filled by the interlayer dielectric layer **121**.

The surface dielectric layer **125**, the interlayer dielectric layer **121**, the passivation layer **132** and the field dielectric layer **134** may include one or more layers of silicon nitride, silicon oxide, silicon oxynitride, aluminum oxide or other atomic layer deposition films, or a multilayer insulator structure, such as an oxide-nitride-oxide layer. In particular embodiments, the surface dielectric layer **125** and the interlayer dielectric layer **121** comprise silicon nitride, the passivation layer **132** comprises silicon oxynitride, and the field dielectric layer **134** comprises silicon nitride.

FIGS. 11A to 11C illustrate GaN HEMT devices according to various further embodiments. For example, FIG. 11A illustrates a HEMT device **100A** that is similar to the HEMT device **100** shown in FIG. 3 (with like reference signs referring to like elements), except that in the device **100A**, the field plate **140** is omitted as discussed above.

FIG. 11B illustrates a HEMT device **100B** that is similar to the HEMT device **100** shown in FIG. 3 (with like reference signs referring to like elements), except that in the HEMT device **100B**, the buried p-type region **113** is omitted as discussed above.

FIG. 11C illustrates a HEMT device **100C** that is similar to the HEMT device **100** shown in FIG. 3 (with like reference signs referring to like elements), except that in the HEMT device **100C**, a second field plate **240** is provided. The second field plate is insulated from the first field plate by the passivation layer **132**. Alternatively, a separate insulating interlayer may be provided between the first field plate **140** and the second field plate **240**. The first and/or second field plates **140**, **240** may each be connected to the source contact **122** or the gate **126**, or may be independently biased.

It will be appreciated that the modifications shown in FIGS. 11A to 11C are not mutually exclusive, and various of the modifications illustrated therein (and others) may be made alone or together to the device shown in FIG. 3.

FIG. 4 illustrates channel layer structures **116A**, **116B** and **116C** of a GaN-based HEMT, such as the HEMT device **100** shown in FIG. 3. The channel layer structures **116A**, **116B** and **116C** have three different types of non-uniform doping profiles according to various embodiments, illustrated as Type A, Type B and Type C. Each channel layer structure **116A-116C** includes three sub-layers **116-1**, **116-2**, **116-3**, where the uppermost sub-layer **116-3** is nearest the barrier layer **118** and farthest from the substrate **112** in the structure shown in FIG. 3. In the examples shown in FIG. 4, the first sub-layer **116-1** is lightly doped and is thicker than the second and third sub-layers **116-2**, **116-3**. For example, the first sub-layer **116-1** in each example may have a thickness

of between 200 and 500 nm, and in particular may have a thickness of about 300 nm. The first sub-layer **116-1** may be doped with silicon dopants at a doping concentration level of less than $3\text{E}16\text{ cm}^{-3}$, and in particular about $1\text{E}16\text{ cm}^{-3}$.

The second and third sub-layers **116-2**, **116-3** in each channel layer structure may have a smaller thickness than the first sub-layers **116-1**. For example, the second and third sub-layers **116-2**, **116-3** may have thicknesses less than 300 nm, such as 50 nm to 150 nm. In particular embodiments, the second and third sub-layers **116-2**, **116-3** may have thicknesses of about 100 nm.

In the Type A channel layer structure **116A**, both the second and third sub-layers **116-2**, **116-3** have increased doping concentration levels relative to the first sub-layer **116-1**. For example, both the second and third sub-layers **116-2**, **116-3** may be doped with silicon at a doping concentration level greater than $1\text{E}16\text{ cm}^{-3}$. For example, both the second and third sub-layers **116-2**, **116-3** may be doped with silicon at a doping concentration level of between about $2\text{E}16\text{ cm}^{-3}$ and $4\text{E}16\text{ cm}^{-3}$. In particular embodiments, both the second and third sub-layers **116-2**, **116-3** may be doped with silicon at a doping concentration level of about $3\text{E}16\text{ cm}^{-3}$.

In the Type B channel layer structure **116B**, the third sub-layer **116-3** has an increased doping concentration level relative to the first sub-layer **116-1** and the second sub-layer **116-2**. For example, the third sub-layer **116-3** may be doped with silicon at a doping concentration level greater than $1\text{E}16\text{ cm}^{-3}$. For example, the third sub-layer **116-3** may be doped with silicon at a doping concentration level of between about $2\text{E}16\text{ cm}^{-3}$ and $4\text{E}16\text{ cm}^{-3}$. In particular embodiments, the third sub-layer **116-3** may be doped with silicon at a doping concentration level of about $3\text{E}16\text{ cm}^{-3}$. The second sub-layer **116-2** may have a doping concentration level less than the doping concentration level of the third sub-layer **116-3**. In particular embodiments, the second layer may have the same doping concentration layer as the first sub-layer **116-1**, e.g., $1\text{E}16\text{ cm}^{-3}$.

In the Type C channel layer structure **116C**, the second sub-layer **116-2** has an increased doping concentration level relative to the first sub-layer **116-1** and the third sub-layer **116-3**. For example, the second sub-layer **116-2** may be doped with silicon at a doping concentration level greater than $1\text{E}16\text{ cm}^{-3}$. For example, the second sub-layer **116-2** may be doped with silicon at a doping concentration level of between about $2\text{E}16\text{ cm}^{-3}$ and $4\text{E}16\text{ cm}^{-3}$. In particular embodiments, the second sub-layer **116-2** may be doped with silicon at a doping concentration level of about $3\text{E}16\text{ cm}^{-3}$. The third sub-layer **116-3** may have a doping concentration level less than the doping concentration level of the second sub-layer **116-2**. In particular embodiments, the third layer may have the same doping concentration layer as the first sub-layer **116-1**, e.g., $1\text{E}16\text{ cm}^{-3}$. Accordingly, in the Type C channel layer structure, the second sub-layer **116-2** forms a buried sub-layer in the channel layer **116** having an increased doping concentration level.

The doping concentration level in the channel layer is preferably kept at a relatively low level, such as below $4\text{E}16\text{ cm}^{-3}$, because increased doping in the channel layer can make it difficult to pinch off the channel, resulting in undesirable current leakage.

FIG. 5A is a graph of simulated values of drain current (I_d) as a function of time following a gate stress event for devices having the three different channel layer structures illustrated in FIG. 4, and FIG. 5B is a graph of simulated values of transconductance (G_m) as a function of gate

voltage (V_g) for devices having the three different channel layer structures illustrated in FIG. 4.

As shown in FIG. 5A, after a -12V gate stress event at time T1, the drain current in the devices with Type A (**502a**) and Type C (**502c**) channel layer structures suffer less I_d drop and recover more quickly than the device with Type B (**502b**) channel layer structure.

As shown in FIG. 5B, the transconductance curve of the devices with Type B (**504b**) and Type C (**504c**) channel layer structures have better transconductance properties (i.e., greater slope) than the device with the Type A (**504a**) channel layer structure. These results illustrate improved device performance for both gate lag and drain lag for the devices with Type C channel layer structure, i.e., devices having buried sub-layers with increased doping concentration levels relative to the upper and lower channel sub-layers.

FIGS. 6A to 6H are cross-sectional views illustrating operations for manufacturing a transistor device including a field plate in accordance with some embodiments.

Referring to FIG. 6A, a silicon carbide substrate **112** is provided on which a channel layer **116** is formed. The channel layer **116** may be doped with carbon and/or iron dopants as is known in the art. An n-type dopant, such as silicon, is incorporated into the channel layer **116** during epitaxial growth to provide a non-uniform doping concentration within the channel layer **116** as described herein in connection with FIG. 4.

A buried p-type region **113** may be formed in the substrate **112** beneath the channel layer and may be connected to the source contact **122** of the device. In various embodiments, the buried p-type region **113** may be floating or may be connected to the gate **126**, or may be independently biased.

The buried p-type region **113** may be formed by implantation of p-type dopants, such as aluminum ions, into the substrate **112** prior to growth of the nitride-based epitaxial layers **116**, **118** thereon.

In some embodiments, the buried p-type region **113** may be formed via epitaxial growth, selective etching and epitaxial regrowth on the substrate and/or via ion implantation. The buried p-type region **113** may extend laterally beneath the source contact **122** and the gate contact **126**. In some embodiments, the buried p-type region **113** may be spaced laterally from the source contact **122** and/or the gate contact **126** so that it does not extend laterally beneath the source contact **122** and/or the gate contact **126**.

Referring to FIG. 6B, a barrier layer **118** is formed on the channel layer **116**, and a preliminary surface dielectric layer **125'** is formed on the barrier layer **118**. A layer of photoresist **52** is formed on the preliminary surface dielectric layer **125'** and patterned to form an opening **54** therein.

The preliminary surface dielectric layer **125'** is selectively etched through the opening **54**, for example using a reactive ion etch or inductively coupled plasma, to form a corresponding opening **154** therein.

Referring to FIG. 6C, a sacrificial dielectric layer **165** is blanket deposited over the preliminary surface dielectric layer **125'** to fill the opening **154**. The sacrificial dielectric layer **165** may be formed of the same material as the preliminary surface dielectric layer **125'**. For example, both the sacrificial dielectric layer **165** and the preliminary surface dielectric layer **125'** may be formed of silicon nitride.

Referring to FIG. 6D, the sacrificial dielectric layer **165** is anisotropically etched, for example using a reactive ion etch or inductively coupled plasma **167**, to remove portions of the sacrificial dielectric layer **165** except for side portions **166** on the inner surfaces of the opening **154** of the preliminary

surface dielectric layer **125'** to form a gate aperture **162** having rounded or beveled edges, as shown in FIG. 6E. The preliminary surface dielectric layer **125'** along with the side portions **166** together form a surface dielectric layer **125** on the barrier layer **118**. With the side portions **166** present, the width of the gate aperture **162** may be about 250 nm.

Referring to FIG. 6F, a metal, such as gold, is deposited and patterned to form a mushroom or T-top gate **126** on the surface dielectric layer **125**. An interlayer dielectric layer **121** is then blanket deposited over the surface dielectric layer **125** and the gate **126**.

Referring to FIG. 6G, a metal, such as gold, is then deposited on the interlayer dielectric layer **121** and patterned to form a field plate **140**. Source **122** and drain **124** contacts are formed on the barrier layer **118**. Heavily doped source and drain contact regions (not shown) may be formed within the barrier layer **118** prior to forming the source **122** and drain **124** contacts thereto to provide ohmic contact to the 2DEG channel.

Referring to FIG. 6H, a passivation layer **132**, such as a layer of SiON, is formed over the field plate **140** and the interlayer dielectric layer **121**. Finally, a field dielectric layer **134**, such as silicon nitride, is formed over the passivation layer **132**.

FIG. 7 is a block diagram illustrating operations of forming a transistor device according to some embodiments. Referring to FIG. 7 and FIGS. 6A to 6H, a method of forming a transistor device according to some embodiments includes providing a channel layer including a sub-layer having an increased doping concentration level relative to a remaining portion of the channel layer (block **802**), forming a barrier layer on the channel layer (block **804**), and forming source and drain contacts on the barrier layer, and a gate contact on the barrier layer between the source and drain contacts (block **806**).

The sub-layer may be a buried sub-layer, the device may further include a second sub-layer between the buried sub-layer and the barrier layer. The buried sub-layer may have a thickness of about 50 nm to 150 nm and a doping concentration level greater than $2 \times 10^{16} \text{ cm}^{-3}$. The buried sub-layer may be doped with an n-type dopant, such as silicon. In some embodiments, the doping concentration level of the n-type dopant may be between about $2 \times 10^{16} \text{ cm}^{-3}$ and $4 \times 10^{16} \text{ cm}^{-3}$.

Transistor devices as described herein may be used in amplifiers that operate in a wide variety of different frequency bands. In some embodiments, the RF transistor amplifiers incorporating transistor devices as described herein may be configured to operate at frequencies greater than 1 GHz. In other embodiments, the RF transistor amplifiers may be configured to operate at frequencies greater than 2.5 GHz. In still other embodiments, the RF transistor amplifiers may be configured to operate at frequencies greater than 3.1 GHz. In yet additional embodiments, the RF transistor amplifiers may be configured to operate at frequencies greater than 5 GHz. In some embodiments, the RF transistor amplifiers may be configured to operate in at least one of the 2.5-2.7 GHz, 3.4-4.2 GHz, 5.1-5.8 GHz, 12-18 GHz, 18-27 GHz, 27-40 GHz or 40-75 GHz frequency bands or sub-portions thereof.

Although embodiments of the inventive concepts have been discussed above with respect to HEMT devices, it will be understood that the inventive concepts described herein may be applied to other types of semiconductor devices, such as MOSFETs, DMOS transistors, and/or laterally diffused MOS (LDMOS) transistors.

RF transistor amplifiers incorporating transistor devices described herein can be used in standalone RF transistor amplifiers and/or in multiple RF transistor amplifiers. Examples of how the RF transistor amplifiers according to some embodiments may be used in applications that include multiple amplifiers will be discussed with reference to FIGS. 8A-8C.

Referring to FIG. 8A, an RF transistor amplifier **1000A** is schematically illustrated that includes a pre-amplifier **1010** and a main amplifier **1030** that are electrically connected in series. As shown in FIG. 8A, RF transistor amplifier **1000A** includes an RF input **1001**, the pre-amplifier **1010**, an inter-stage impedance matching network **1020**, the main amplifier **1030**, and an RF output **1002**. The inter-stage impedance matching network **1020** may include, for example, inductors and/or capacitors arranged in any appropriate configuration in order to form a circuit that improves the impedance match between the output of pre-amplifier **1010** and the input of main amplifier **1030**. While not shown in FIG. 8A, RF transistor amplifier **1000A** may further include an input matching network that is interposed between RF input **1001** and pre-amplifier **1010**, and/or an output matching network that is interposed between the main amplifier **1030** and the RF output **1002**. The RF transistor amplifiers according to embodiments may be used to implement either or both of the pre-amplifier **1010** and the main amplifier **1030**.

Referring to FIG. 8B, an RF transistor amplifier **1000B** is schematically illustrated that includes an RF input **1001**, a pair of pre-amplifiers **1010-1**, **1010-2**, a pair of inter-stage impedance matching networks **1020-1**, **1020-2**, a pair of main amplifiers **1030-1**, **1030-2**, and an RF output **1002**. A splitter **1003** and a combiner **1004** are also provided. Pre-amplifier **1010-1** and main amplifier **1030-1** (which are electrically connected in series) are arranged electrically in parallel with pre-amplifier **1010-2** and main amplifier **1030-2** (which are electrically connected in series). As with the RF transistor amplifier **1000A** of FIG. 9A, RF transistor amplifier **1000B** may further include an input matching network that is interposed between RF input **1001** and pre-amplifiers **1010-1**, **1010-2**, and/or an output matching network that is interposed between the main amplifiers **1030-1**, **1030-2** and the RF output **1002**.

As shown in FIG. 8C, the RF transistor amplifiers according to some embodiments may also be used to implement Doherty amplifiers. As is known in the art, a Doherty amplifier circuit includes first and second (or more) power-combined amplifiers. The first amplifier is referred to as the "main" or "carrier" amplifier and the second amplifier is referred to as the "peaking" amplifier. The two amplifiers may be biased differently. For example, the main amplifier may comprise a Class AB or a Class B amplifier while the peaking amplifier may be a Class C amplifier in one common Doherty amplifier implementation. The Doherty amplifier may operate more efficiently than balanced amplifiers when operating at power levels that are backed off from saturation. An RF signal input to a Doherty amplifier is split (e.g., using a quadrature coupler), and the outputs of the two amplifiers are combined. The main amplifier is configured to turn on first (i.e., at lower input power levels) and hence only the main amplifier will operate at lower power levels. As the input power level is increased towards saturation, the peaking amplifier turns on and the input RF signal is split between the main and peaking amplifiers.

As shown in FIG. 8C, the Doherty RF transistor amplifier **1000C** includes an RF input **1001**, an input splitter **1003**, a main amplifier **1040**, a peaking amplifier **1050**, an output

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combiner **1004** and an RF output **1002**. The Doherty RF transistor amplifier **1000C** includes a 90° transformer **1007** at the input of the peaking amplifier **1050** and a 90° transformer **1005** at the input of the main amplifier **1040**, and may optionally include input matching networks and/or an output matching networks (not shown). The main amplifier **1040** and/or the peaking amplifier **1050** may be implemented using any of the above-described RF transistor amplifiers according to embodiments.

The RF transistor amplifiers according to embodiments may be formed as discrete devices, or may be formed as part of a Monolithic Microwave Integrated Circuit (MMIC). A MMIC refers to an integrated circuit that operates on radio and/or microwave frequency signals in which all of the circuitry for a particular function is integrated into a single semiconductor chip. An example MMIC device is a transistor amplifier that includes associated matching circuits, feed networks and the like that are all implemented on a common substrate. MMIC transistor amplifiers typically include a plurality of unit cell HEMT transistors that are connected in parallel.

FIG. 9 is a plan view of a MMIC RF transistor amplifier **400** according to embodiments of the present inventive concepts. As shown in FIG. 9, the MMIC RF transistor amplifier **400** includes an integrated circuit chip **430** that is contained within a package **410**. The package **410** may comprise a protective housing that surrounds and protects the integrated circuit chip **430**. The package **410** may be formed of, for example, a ceramic material.

The package **410** includes an input lead **412** and an output lead **418**. The input lead **412** may be mounted to an input lead pad **414** by, for example, soldering. One or more input bond wires **420** may electrically connect the input lead pad **414** to an input bond pad on the integrated circuit chip **430**. The integrated circuit chip **430** includes an input feed network **438**, an input impedance matching network **450**, a first RF transistor amplifier stage **460**, an intermediate impedance matching network **440**, a second RF transistor amplifier stage **462**, an output impedance matching stage **470**, and an output feed network **482**.

The package **410** further includes an output lead **418** that is connected to an output lead pad **416** by, for example, soldering. One or more output bond wires **490** may electrically connect the output lead pad **416** to an output bond pad on the integrated circuit chip **430**. The first RF transistor amplifier stage **460** and/or the second RF transistor amplifier stage **462** may be implemented using any of the RF transistor amplifiers according to embodiments of the present inventive concepts.

The RF transistor amplifiers according to embodiments of the present inventive concepts may be designed to operate in a wide variety of different frequency bands. In some embodiments, these RF transistor amplifier dies may be configured to operate in at least one of the 0.6-2.7 GHz, 3.4-4.2 GHz, 5.1-5.8 GHz, 12-18 GHz, 18-27 GHz, 27-40 GHz or 40-75 GHz frequency bands or sub-portions thereof. The techniques according to embodiments of the present inventive concepts may be particularly advantageous for RF transistor amplifiers that operate at frequencies of 10 GHz and higher.

FIGS. 10A and 10B are schematic cross-sectional views illustrating several example ways that the RF transistor amplifier dies according to embodiments of the present inventive concepts may be packaged to provide packaged RF transistor amplifiers **600A** and **600B**, respectively.

FIG. 10A is a schematic side view of a packaged Group III nitride-based RF transistor amplifier **600A**. As shown in

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FIG. 10A, packaged RF transistor amplifier **600A** includes the RF transistor amplifier die **100** packaged in an open cavity package **610A**. The package **610A** includes metal gate leads **622A**, metal drain leads **624A**, a metal submount **630**, sidewalls **640** and a lid **642**.

The submount **630** may include materials configured to assist with the thermal management of the package **600A**. For example, the submount **630** may include copper and/or molybdenum. In some embodiments, the submount **630** may be composed of multiple layers and/or contain vias/interconnects. In an example embodiment, the submount **630** may be a multilayer copper/molybdenum/copper metal flange that comprises a core molybdenum layer with copper cladding layers on either major surface thereof. In some embodiments, the submount **630** may include a metal heat sink that is part of a lead frame or metal slug. The sidewalls **640** and/or lid **642** may be formed of or include an insulating material in some embodiments. For example, the sidewalls **640** and/or lid **642** may be formed of or include ceramic materials.

In some embodiments, the sidewalls **640** and/or lid **642** may be formed of, for example, Al₂O₃. The lid **642** may be glued to the sidewalls **640** using an epoxy glue. The sidewalls **640** may be attached to the submount **630** via, for example, braising. The gate lead **622A** and the drain lead **624A** may be configured to extend through the sidewalls **640**, though embodiments of the present inventive concepts are not limited thereto.

The RF transistor amplifier die **100** is mounted on the upper surface of the metal submount **630** in an air-filled cavity **612** defined by the metal submount **630**, the ceramic sidewalls **640** and the ceramic lid **642**. The gate and drain terminals of RF transistor amplifier die **100** may be on the top side of the structure, while the source terminal is on the bottom side of the structure.

The gate lead **622A** may be connected to the gate terminal of RF transistor amplifier die **100** by one or more bond wires **654**. Similarly, the drain lead **624A** may be connected to the drain terminal of RF transistor amplifier die **100** by one or more bond wires **654**. The source terminal may be mounted on the metal submount **630** using, for example, a conductive die attach material (not shown). The metal submount **630** may provide the electrical connection to the source terminal and may also serve as a heat dissipation structure that dissipates heat that is generated in the RF transistor amplifier die **100**.

The heat is primarily generated in the upper portion of the RF transistor amplifier die **100** where relatively high current densities are generated in, for example, the channel regions of the unit cell transistors **102**. This heat may be transferred through the source vias **146** and the semiconductor layer structure **150** to the source terminal and then to the metal submount **630**.

FIG. 10B is a schematic side view of another packaged Group III nitride based RF transistor amplifier **600B**. RF transistor amplifier **600B** differs from RF transistor amplifier **600A** in that it includes a different package **610B**. The package **610B** includes a metal submount **630**, as well as metal gate and drain leads **622B**, **624B**. RF transistor amplifier **600B** also includes a plastic overmold **660** that at least partially surrounds the RF transistor amplifier die **100**, the leads **622B**, **624B**, and the metal submount **630**.

Other components of RF transistor amplifier **600B** may be the same as the like-numbered components of RF transistor amplifier **600A** and hence further description thereof will be omitted. While embodiments of the present inventive concepts are described above with respect to gallium nitride

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based RF transistor amplifiers, it will be appreciated that embodiments of the inventive concepts are not limited thereto. For example, the transistors described above may also be used as power transistors in switching and other applications.

Embodiments of the present inventive concepts have been described above with reference to the accompanying drawings, in which embodiments of the inventive concepts are shown. The inventive concepts may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the inventive concepts to those skilled in the art. Like numbers refer to like elements throughout. In the specification and the figures, two-part reference numbers (i.e., two numbers separated by a dash, such as 100-1) may be used to identify like elements. When such two-part reference numbers are employed, the full reference numeral may be used to refer to a specific instance of the element, while the first part of the reference numeral may be used to refer to the elements collectively.

It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present inventive concepts. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the inventive concepts. As used herein, the terms “comprises,” “comprising,” “includes” and/or “including” specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

It will be understood that when an element such as a layer, region or substrate is referred to as being “on” or extending “onto” another element, it can be directly on or extend directly onto the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly on” or extending “directly onto” another element, there are no intervening elements present. It will also be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present. Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “lateral” or “vertical” may be used herein to describe a relationship of one element, layer or region to another element, layer or region as illustrated in the figures.

Transistor structures with features that may be used in embodiments of the present inventive concepts are disclosed in the following commonly assigned publications, the contents of each of which are fully incorporated by reference herein in their entirety: U.S. Pat. No. 6,849,882 to Chavarkar et al. and entitled “Group-III Nitride Based High Electron Mobility Transistor (HEMT) With Barrier/Spacer Layer”; U.S. Pat. No. 7,230,284 to Parikh et al. and entitled “Insulating Gate AlGaIn/GaN HEMT”; U.S. Pat. No. 7,501,669 to

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Parikh et al. and entitled “Wide Bandgap Transistor Devices With Field Plates”; U.S. Pat. No. 7,126,426 to Mishra et al. and entitled “Cascode Amplifier Structures Including Wide Bandgap Field Effect Transistor With Field Plates”; U.S. Pat. No. 7,550,783 to Wu et al. and entitled “Wide Bandgap HEMTs With Source Connected Field Plates”; U.S. Pat. No. 7,573,078 to Wu et al. and entitled “Wide Bandgap Transistors With Multiple Field Plates”; U.S. Pat. Pub. No. 2005/0253167 to Wu et al. and entitled “Wide Bandgap Field Effect Transistors With Source Connected Field Plates”; U.S. Pat. Pub. No. 2006/0202272 to Wu et al. and entitled “Wide Bandgap Transistors With Gate-Source Field Plates”; U.S. Pat. Pub. No. 2008/0128752 to Wu and entitled “GaN Based HEMTs With Buried Field Plates”; U.S. Pat. Pub. No. 2010/0276698 to Moore et al. and entitled “Gate Electrodes For Millimeter-Wave Operation and Methods of Fabrication”; U.S. Pat. Pub. No. 2012/0049973 to Smith, Jr. et al. and entitled “High Power Gallium Nitride Field Effect Transistor Switches”; U.S. Pat. Pub. No. 2012/0194276 to Fisher and entitled “Low Noise Amplifiers Including Group III Nitride Based High Electron Mobility Transistors”; and U.S. Pat. No. 9,847,411 to Sriram et al. entitled “Recessed field plate transistor structures.”

Although embodiments of the inventive concepts have been described in considerable detail with reference to certain configurations thereof, other versions are possible. The field plates and gates can also have many different shapes and can be connected to the source contact in many different ways. Accordingly, the spirit and scope of the inventive concepts should not be limited to the specific embodiments described above.

What is claimed is:

1. A transistor device, comprising:

a substrate;
a channel layer formed on the substrate;
a barrier layer on the channel layer; and
a source contact and a drain contact on the barrier layer, and a gate contact that terminates on a surface of the barrier layer between the source and drain contacts;
wherein the channel layer comprises a first sub-layer, a second sub-layer on the first sub-layer, and a third sub-layer on the second sub-layer, wherein the first sub-layer is between the second sub-layer and the substrate and the second sub-layer is between the first sub-layer and the third sub-layer;
wherein the first sub-layer is doped with n-type dopants at a first doping concentration, the second sub-layer is doped with n-type dopants at a second doping concentration, and the third sub-layer is doped with n-type dopants at a third doping concentration, wherein the second doping concentration is greater than the first doping concentration.

2. The transistor device of claim 1, wherein the second sub-layer has a thickness between 50 and 150 nm.

3. The transistor device of claim 2, wherein the second sub-layer has a thickness of about 100 nm.

4. The transistor device of claim 1, wherein the first sub-layer has a thickness greater than the second sub-layer.

5. The transistor device of claim 4, wherein the first sub-layer has a thickness of about 100 nm.

6. The transistor device of claim 1, wherein the second sub-layer is doped with an n-type dopant having a doping concentration level greater than $2 \times 10^{16} \text{ cm}^{-3}$.

7. The transistor device of claim 6, wherein the first sub-layer is doped with an n-type dopant having a doping concentration level less than $2 \times 10^{16} \text{ cm}^{-3}$.

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8. The transistor device of claim 1, wherein the n-type dopant in the second sub-layer comprises silicon.

9. The transistor device of claim 1, wherein the doping concentration level of the n-type dopant in the second sub-layer is between about $2 \times 10^{16} \text{ cm}^{-3}$ and $4 \times 10^{16} \text{ cm}^{-3}$. 5

10. The transistor device of claim 1, wherein the doping concentration level of the n-type dopant in the second sub-layer is about $3 \times 10^{16} \text{ cm}^{-3}$.

11. The transistor device of claim 1, wherein the doping concentration level of the n-type dopant in the first sub-layer is about $1 \times 10^{16} \text{ cm}^{-3}$. 10

12. The transistor device of claim 1, wherein the doping concentration level of the n-type dopant in the first sub-layer is below about $2 \times 10^{16} \text{ cm}^{-3}$.

13. The transistor device of claim 1, wherein the channel layer has a total thickness of about 500 nm or less. 15

14. The transistor device of claim 1, wherein the barrier layer and the channel layer cooperate to induce a two-dimensional electron gas (2DEG) in the channel layer.

15. The transistor device of claim 1, wherein the channel layer comprises GaN. 20

16. The transistor device of claim 1, wherein the third doping concentration is lower than the second doping concentration.

17. The transistor device of claim 16, wherein the first sub-layer has a thickness greater than a thickness of the second sub-layer and greater than a thickness of the third sub-layer. 25

18. A transistor device, comprising:
a substrate;
a channel layer on the substrate;
a barrier layer on the channel layer; and

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source and drain contacts on the barrier layer, and a gate contact that terminates on a surface of the barrier layer between the source and drain contacts;

wherein the channel layer comprises a first sub-layer having a first doping concentration level, a second sub-layer having a second doping concentration level on the first sub-layer, and a third sub-layer having a third doping concentration level on the second sub-layer opposite the first sub-layer, wherein the third sub-layer is adjacent the barrier layer, and wherein the first sub-layer is between the second sub-layer and the substrate, wherein each of the first sub-layer, the second sub-layer, and the third sub-layer is doped with an n-type dopant;

wherein the second doping concentration level is higher than the first doping concentration level and the third doping concentration level,

wherein the second doping concentration level is greater than $2 \times 10^{16} \text{ cm}^{-3}$, and

wherein the first and third doping concentration levels are each less than about $2 \times 10^{16} \text{ cm}^{-3}$.

19. The transistor device of claim 18, wherein the second doping concentration level is between about $2 \times 10^{16} \text{ cm}^{-3}$ and $4 \times 10^{16} \text{ cm}^{-3}$.

20. The transistor device of claim 19, wherein the first and third doping concentration levels are each about $1 \times 10^{16} \text{ cm}^{-3}$, and wherein the second doping concentration level is about $3 \times 10^{16} \text{ cm}^{-3}$. 25

21. The transistor device of claim 18, wherein the first and second sub-layers each have a thickness between about 50 nm and 150 nm. 30

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